

Logic Building Assignment : 22

Create separate visual Studio project for each problem statement separately.

1. Accept N numbers from user and return frequency of even numbers.

Input : N : 6

Elements : 85 66 3 80 93 88

Output : 3

Program Layout :

```
#include<stdio.h>

int CountEven(int Arr[], int iLength)
{
    // Logic
}

int main()
{
    int iSize = 0,iRet = 0,iCnt = 0;
    int *p = NULL;

    printf("Enter number of elements");
    scanf("%d",&iSize);

    p = (int *)malloc(iSize * sizeof(int));

    if(p == NULL)
    {
        printf("Unable to allocate memory");
        return -1;
    }

    printf("Enter %d elements ",iLength);

    for(iCnt = 0;i<iLength; iCnt++)
    {
        printf("Enter element : %d",iCnt+1);
        scanf("%d",&p[iCnt]);
    }

    iRet = CountEven(p, iSize);
```

```
printf("Result is %d",iRet);

free(p);

return 0;
}
```

2. Accept N numbers from user and return difference between frequency of even number and odd numbers.

Input : N : 7

 Elements : 85 66 3 80 93 88 90

Output : 1 (4 -3)

Program Layout :

```
#include<stdio.h>

Int Frequency(int Arr[], int iLength)
{
    // Logic
}

int main()
{
    int iSize = 0,iRet = 0,iCnt = 0, iRet = 0;
    int *p = NULL;

    printf("Enter number of elements");
    scanf("%d",&iSize);

    p = (int *)malloc(iSize * sizeof(int));

    if(p == NULL)
    {
        printf("Unable to allocate memory");
        return -1;
    }

    printf("Enter %d elements ",iLength);

    for(iCnt = 0;iCnt<iLength; iCnt++)
    {
        printf("Enter element : %d",iCnt+1);
```

```

        scanf("%d",&p[iCnt]);
    }

    iRet = Frequency(p, iSize);

    printf("%d",iRet);

    free(p);

    return 0;
}

```

3. Accept N numbers from user check whether that numbers contains 11 in it or not.

Input : N : 6

 Elements : 85 66 11 80 93 88

Output : 11 is present

Input : N : 6

 Elements : 85 66 3 80 93 88

Output : 11 is absent

Program Layout :

```
#include<stdio.h>
```

```

bool Check(int Arr[], int iLength)
{
    // Logic
}

```

```

int main()
{
    int iSize = 0,iRet = 0,iCnt = 0;
    int *p = NULL;
    bool bRet = false;

    printf("Enter number of elements");
    scanf("%d",&iSize);

    p = (int *)malloc(iSize * sizeof(int));
}

```

```
if(p == NULL)
{
    printf("Unable to allocate memory");
    return -1;
}

printf("Enter %d elements ",iLength);

for(iCnt = 0;iCnt<iLength; iCnt++)
{
    printf("Enter element : %d",iCnt+1);
    scanf("%d",&p[iCnt]);
}

bRet = Check(p, iSize);

if(bRet == true)
{
    printf("11 is present");
}
else
{
    printf("11 is absent");
}
free(p);
return 0;
}
```

4. Accept N numbers from user and return frequency of 11 form it.

Input : N : 6

 Elements : 85 66 3 15 93 88

Output : 0

Input : N : 6

 Elements : 85 11 3 15 11 111

Output : 2

Program Layout :

```
#include<stdio.h>
```

```
int Frequency(int Arr[], int iLength)
{
    // Logic
}

int main()
{
    int iSize = 0,iRet = 0,iCnt = 0, iRet = 0;
    int *p = NULL;

    printf("Enter number of elements");
    scanf("%d",&iSize);

    p = (int *)malloc(iSize * sizeof(int));

    if(p == NULL)
    {
        printf("Unable to allocate memory");
        return -1;
    }

    printf("Enter %d elements ",iLength);

    for(iCnt = 0;iCnt<iLength; iCnt++)
    {
        printf("Enter element : %d",iCnt+1);
        scanf("%d",&p[iCnt]);
    }

    iRet = Frequency(p, iSize);

    printf("%d",iRet);

    free(p);

    return 0;
}
```

5. Accept N numbers from user and accept one another number as NO , return frequency of NO form it.

Input : N : 6

 NO: 66

 Elements : 85 66 3 66 93 88

Output : 2

Input : N : 6

NO: 12

Elements : 85 11 3 15 11 111

Output : 0

Program Layout :

```
#include<stdio.h>
```

```
int Frequency(int Arr[], int iLength, int iNo)
{
    // Logic
}
```

```
int main()
{
    int iSize = 0,iRet = 0,iCnt = 0, iRet = 0, iValue = 0;
    int *p = NULL;
```

```
    printf("Enter number of elements");
    scanf("%d",&iSize);
```

```
    printf("Enter the number");
    scanf("%d",&iValue);
```

```
    p = (int *)malloc(iSize * sizeof(int));
```

```
    if(p == NULL)
    {
        printf("Unable to allocate memory");
        return -1;
    }
```

```
    printf("Enter %d elements ",iLength);
```

```
    for(iCnt = 0;iCnt<iLength; iCnt++)
    {
        printf("Enter element : %d",iCnt+1);
        scanf("%d",&p[iCnt]);
    }
```

```
iRet = Frequency(p, iSize,iValue);  
  
printf("%d",iRet);  
  
free(p);  
  
return 0;  
}
```

